

Transverse UV, and the $y' \rightarrow z$ region

Which divergence actually blocks the $+$ prescription

$$r \equiv y' - z, \quad \xi \equiv p^+ / k^+, \quad \text{ph} \equiv e^{-i\xi k \cdot (y' - z)}, \quad \ell_k \equiv \log \frac{k^+}{\Lambda}$$

0. The short version

There is no transverse ultraviolet divergence — not at $y' \rightarrow z$, not anywhere. Your criterion is the right one, and at short distance it is satisfied.

But $y' \rightarrow z$ is still the region that matters, for a different reason: it is where the phase stops depending on p^+ , so it is the *source* of the rapidity divergence. That is a longitudinal divergence living in a transverse region, not a transverse divergence — and the $+$ prescription handles it beautifully there (§2).

What blocks the split is at the other end: $|y' - z| \rightarrow \infty$. That divergence is *infrared*, not ultraviolet — and it blocks the split just as effectively, because the criterion is “no transverse divergence”, not “no transverse UV divergence”.

1. $y' \rightarrow z$: no ultraviolet divergence

At $y' = z$ only **one** of the four kernels is singular, $\frac{(y'-z)^m}{(y'-z)^2}$; the other three ($x' - y'$, $y - w$, $x - z$) are regular there. So the integrand is $O(1/\rho)$ and $\int \rho d\rho \rho^{-1} = \int d\rho$ converges. In fact it does better: the leading term integrates to *zero* by parity, $\int_{|r|<\epsilon} d^2 r \frac{r^m}{r^2} = 0$, so the small-disc integral is $O(\epsilon^2)$. Measured $|\int_{|y'-z|<\epsilon} d^2 y'|$ for each index structure:

structure	eps=1e-01	eps=1e-02	eps=1e-03	ratios/decade	
T1 x ph	3.386e-04	3.386e-06	3.386e-08	0.0100	0.0100
T2 x ph	3.938e-03	3.938e-05	3.938e-07	0.0100	0.0100
T3 x ph	1.178e-02	1.178e-04	1.178e-06	0.0100	0.0100
T2 x (ph-1)	5.900e-04	5.870e-06	5.870e-08	0.0099	0.0100
T2 bare (g(0))	3.894e-03	3.894e-05	3.894e-07	0.0100	0.0100

A ratio of 0.01 per decade is $F \propto \epsilon^2$, for every structure — with the phase, without it, and for the subtracted combination alike. **No UV divergence at $y' \rightarrow z$.**

And none from the points collapsing together

The only way to build a ρ^{-2} is to make two kernels singular at once, which needs two points to collide — and that costs measure. Counting relative coordinates (two-dimensional each) against singular kernels:

collapsing points	rel. coords	measure	kernels singular	integrand	net
$y' \sim z$	1	$\rho^2 \frac{d\rho}{\rho}$	1	ρ^{-1}	$\rho^1 \frac{d\rho}{\rho}$
$y' \sim z \sim x'$	2	$\rho^4 \frac{d\rho}{\rho}$	2	ρ^{-2}	$\rho^2 \frac{d\rho}{\rho}$
$y' \sim z \sim x$	2	$\rho^4 \frac{d\rho}{\rho}$	2	ρ^{-2}	$\rho^2 \frac{d\rho}{\rho}$
$y' \sim z \sim x' \sim x$	3	$\rho^6 \frac{d\rho}{\rho}$	3	ρ^{-3}	$\rho^3 \frac{d\rho}{\rho}$

Every net power is positive. The transverse integrations are ultraviolet finite, and the $+$ prescription cannot change that — the phase is analytic at short distance.

2. Why $y' \rightarrow z$ is nevertheless the crucial region

It is where $\text{ph} = e^{-i\xi k \cdot (y' - z)} \rightarrow 1$, i.e. where the integrand stops knowing about p^+ . The $\int \frac{d\xi}{\xi}$ is then completely unprotected, and that is exactly where the rapidity logarithm is generated. **A longitudinal divergence localised in a transverse region is not a transverse divergence** — and this one is what the $+$ prescription is built for.

Indeed the subtraction is doubly effective there. Expanding,

$$\text{ph} - 1 = -i\xi k \cdot (y' - z) + O(\rho^2),$$

so it supplies a factor ξ — cancelling the $\frac{1}{\xi}$ pole — *and* a factor ρ , softening the kernel from ρ^{-1} to $O(1)$. The numbers show it: the subtracted disc integral is $6.7\times$ smaller than the unsubtracted one at every ϵ ,

eps=1e-02:	T2 x ph	3.938e-05	T2 x (ph-1)	5.870e-06
eps=1e-03:	T2 x ph	3.938e-07	T2 x (ph-1)	5.870e-08
eps=1e-04:	T2 x ph	3.938e-09	T2 x (ph-1)	5.870e-10

So at $y' \rightarrow z$ your reasoning is exactly right: no transverse divergence, and the $+$ prescription applies cleanly.

3. The obstruction is at the far end, and it is infrared

The region that fails is $|y' - z| \rightarrow \infty$, where the phase is the *only* thing suppressing the integrand and the subtraction removes it. Restating the measurement from the previous note, $dF/d\log R$ for $F(R) = \int_{|z|<R} d^2z$:

structure	$\xi = 0.3$	$\xi = 0.05$
$S_2 \times \text{ph}$ (before splitting)	-0.00022	+0.00005
$S_2 \times (\text{ph} - 1)$ (the $+$ term)	-6.28352	-6.28314
S_2 bare = $g(0)$ (the log term)	+6.28319	+6.28319

against $-2\pi = -6.28319$. So the criterion you are applying is the right one — it is just not satisfied, and the failure is at large distance rather than small.

Why this is structural, not accidental. In a dipole calculation the large- z convergence is normally supplied by the *kernel*: the full BK kernel $\mathcal{M} = \frac{(x-y')^2}{(x-z)^2(y'-z)^2}$ falls as ρ^{-4} . Short-distance $z \rightarrow x$, $z \rightarrow y'$ is handled by the *colour* structure vanishing there. Here the short distance is fine for the reason in §1 — each kernel is single — but you have only the real piece $-2\mathcal{K} \sim \rho^{-2}$, so the large- z end has nothing holding it up. Nor does the colour structure help: as $|z| \rightarrow \infty$, $U(z) \rightarrow 1$ and the two brackets become

$$\delta^{bd'} [U^{e'e'}(y') - U^{e'e'}(x')] \rho^{e'}(x'), \quad U^{ad}(y) \rho^d(y) [U^{be}(x) \rho^e(x) - \rho^b(x)],$$

which are z -independent, not zero. The coefficient of the logarithm survives.

4. On rewriting the logarithm

Yes, $\log \frac{\vee - k^+}{\Lambda} = \log \frac{\vee}{\Lambda} + \log(1 - \frac{k^+}{\vee})$ is exact, and the second piece is finite (and vanishes as $\vee \rightarrow \infty$). Same for splitting it as $\ell_k + \log \frac{\vee - k^+}{k^+}$. All of that is legitimate bookkeeping.

It does not help with the divergence, though, because **the divergent object is the coefficient, not the logarithm**. The second term is (transversely divergent $g(0)$) $\times \log[\dots]$; rewriting the log as a sum of logs just distributes the same infinite coefficient over more terms. Only making $g(0)$ convergent does anything — and that means the virtual diagrams, which turn $-2\mathcal{K}$ into $\mathcal{M}$ and ρ^{-2} into ρ^{-4} .

Where the rewriting *does* earn its keep is afterwards: once $g(0)$ is finite, choosing ℓ_k (via a $\theta(k^+ - p^+)$ subtraction) rather than $\log \frac{\vee - k^+}{\Lambda}$ gives each piece a finite $\vee \rightarrow \infty$ limit and matches the convention BK evolution expects.

zint/src/uv_y pz.py — the short-distance numbers, written to zint/RESULTS_uvypz.txt; zint/src/zdiv_direct.py — the large-distance slopes.